

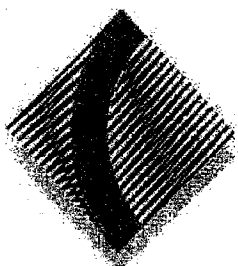
AT

Name: _____

Class: 12MTZ1

Teacher: MR TONG

CHERRYBROOK TECHNOLOGY HIGH SCHOOL



2010 AP4

YEAR 12 TRIAL HSC EXAMINATION

MATHEMATICS EXTENSION 2

*Time allowed - 3 HOURS
(Plus 5 minutes' reading time)*

DIRECTIONS TO CANDIDATES:

- Attempt all questions.
- All questions are of equal value.
- Each question is to be commenced on a new page clearly marked Question 1, Question 2, etc on the top of the page. **
- All necessary working should be shown in every question. Full marks may not be awarded for careless or badly arranged work.
- Approved calculators may be used. Standard Integral Tables are provided.
- Your solutions will be collected in one bundle stapled in the top left corner.

Please arrange them in order, Q1 to 8.

****Each page must show your name and your class. ****

Total Marks – 120
Attempt Questions 1–8
All questions are of equal value

Question 1 (15 marks) Start a NEW PAGE.

MARKS

a) Find $\int \frac{x^2}{\sqrt{x^3 - 1}} dx$ 2

b) Find $\int \frac{dx}{\sqrt{8x^2 + 2}}$ 2

c) Evaluate $\int_0^{\frac{1}{2}} \sin^{-1} x dx$ 3

d) (i) Find A and B if $\frac{x^2 - 4x + 2}{(2x + 1)(x^2 + 4)} = \frac{1}{2x + 1} + \frac{Ax + B}{x^2 + 4}$ 2

(ii) Hence evaluate $\int_0^2 \frac{x^2 - 4x + 2}{(2x + 1)(x^2 + 4)} dx$, leave your answer in simplest exact form. 2

e) Evaluate $\int_0^{\frac{\pi}{2}} \frac{d\theta}{1 + \sin \theta + \cos \theta}$ 4

Question 2 (15 marks) Start a NEW PAGE.

MARKS

a) Let $z = 5 - 12i$, $w = 3 + 4i$. Find, in the form of $a + bi$

(i) $z + \overline{w}$

1

(ii) $\frac{z}{w}$

2

b) (i) Show that $\tan \frac{\pi}{12} = \frac{\sqrt{3} - 1}{\sqrt{3} + 1}$.

1

(ii) Express $z = (\sqrt{3} + 1) + (\sqrt{3} - 1)i$ in modulus argument form.

2

(iii) Find the least positive integer n for which z^n is a real number.

1

c) (i) Shade the region $|z - 3 - 8i| \leq 3$,

2

(ii) Find the minimum value of $|z + 3|$ where z satisfies the condition in (i).

2

d) Sketch in the complex plane, the locus of

(i) $\arg z = \frac{\pi}{4}$

1

(ii) $\arg(\overline{z}) = \frac{\pi}{4}$.

1

e) On the Argand diagram, OA represents the complex number $z_1 = x + iy$,

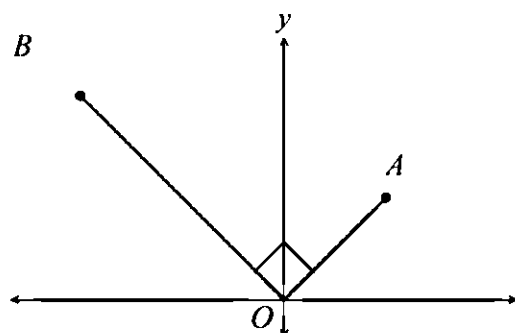
$\angle AOB = \frac{\pi}{2}$ and the length of OB is twice that of OA.

(i) Show that OB represents the complex number $-2y + 2ix$.

1

(ii) Given that AOB is a rectangle, find the complex number represented by OC.

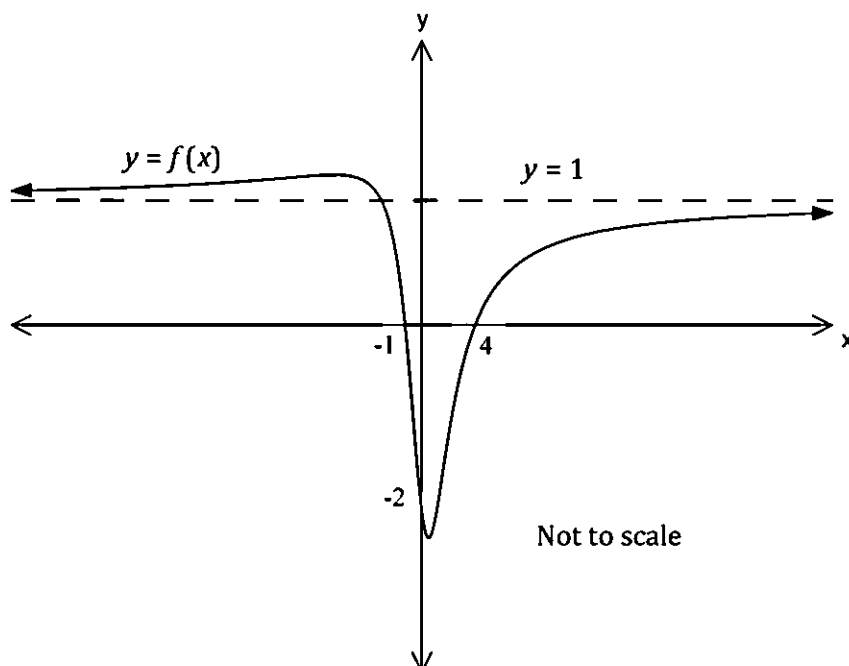
1



Question 3 (15 marks) Start a NEW PAGE.

MARKS

a)



The diagram shows the graph of $y = f(x)$. $y = 1$ is the horizontal asymptote. The graph intersects the x -axis at the points $(-1, 0)$ and $(4, 0)$, and the y -axis at the point $(0, -2)$. Detach the printed graphs from the end of the question booklet and sketch on them the graphs of the following, without using calculus, indicating any asymptotes.

(i) $y = \sqrt{f(x)}$ 2

(ii) $y = \frac{1}{f(x)}$ 2

(iii) $y = f(|x|)$ 2

b) $P(x) = x^4 - 2x^3 + 3x^2 - 4x + 1$ and the equation $P(x) = 0$ has roots α , β , γ and δ .

(i) Show that the equation $P(x) = 0$ has no integer roots. 1

(ii) Show that $P(x) = 0$ has a real root between 0 and 1. 1

(iii) Evaluate $\alpha^2 + \beta^2 + \gamma^2 + \delta^2$. 2

(iv) Hence deduce that the $P(x) = 0$ has 2 real roots. 2

(v) Find the equation whose roots are $\frac{1}{2\alpha\beta\gamma}$, $\frac{1}{2\alpha\beta\delta}$, $\frac{1}{2\alpha\gamma\delta}$ and $\frac{1}{2\beta\gamma\delta}$. 2

(vi) Hence evaluate $\frac{1}{2\alpha\beta\gamma} + \frac{1}{2\alpha\beta\delta} + \frac{1}{2\alpha\gamma\delta} + \frac{1}{2\beta\gamma\delta}$. 1

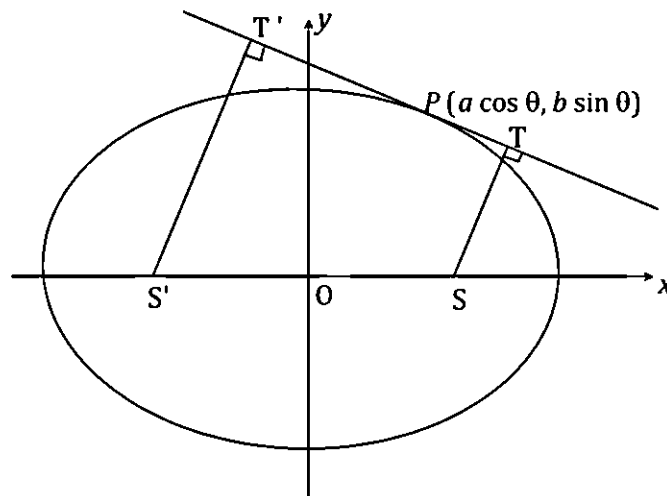
Question 4 (15 marks) Start a NEW PAGE.

MARKS

a) The equation $|z - 2 + i| + |z - 10 + i| = 12$ corresponds to an ellipse on the Argand diagram.

- | | | |
|------|---|---|
| (i) | Write down the complex number corresponding to the centre of the ellipse. | 1 |
| (ii) | Sketch the ellipse and state the lengths of the major and minor axes. | 3 |

b)



The point $P(a \cos \theta, b \sin \theta)$ lies on the ellipse: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. The points T and T' are the feet of the perpendiculars from the foci S and S' respectively to this tangent.

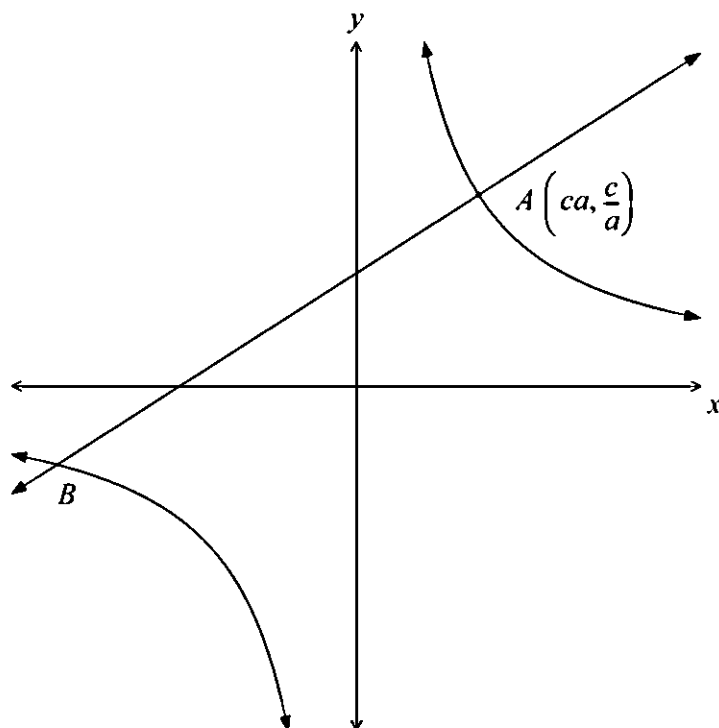
- | | | |
|------|--|---|
| (i) | Write down the equation of the tangent at P . | 1 |
| (ii) | Show that $ST = \frac{ab e \cos \theta - 1 }{\sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}}$. | 2 |
| (ii) | Hence prove that $ST \times S'T' = b^2$. | 2 |

Question 4 is continued on page 5.

Question 4 (continued)

MARKS

c)



The point $A \left(ca, \frac{c}{a} \right)$, where $a \neq \pm 1$ lies on the hyperbola $xy = c^2$. The normal through A meets the other branch of the curve at B .

- (i) Show that the equation of the normal through A is 2

$$y = a^2x + \frac{c}{a}(1 - a^4)$$

- (ii) Hence if B has coordinates $\left(cb, \frac{c}{b} \right)$, show that $b = \frac{-1}{a^3}$. 1

- (iii) $M(x, y)$ is the mid-point of AB . Show that at M , $a^2 = -\frac{y}{x}$. 1

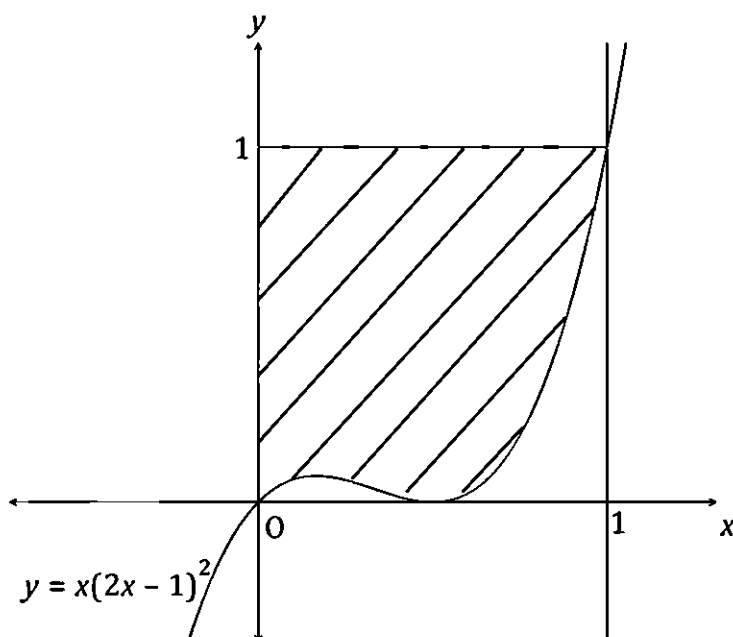
- (iv) Hence show that the equation of the locus of M is $4x^3y^3 + c^2(x^2 - y^2)^2 = 0$. 2

Question 5 (15 marks) Start a NEW PAGE.

MARKS

a)

4



A solid is formed by rotating the shaded region bounded by the curve $y = x(2x - 1)^2$, the y -axis and the line $y = 1$ about the y -axis. Use the method of cylindrical shells to find the volume of this solid.

b) If $z = \cos \theta + i \sin \theta$, show that

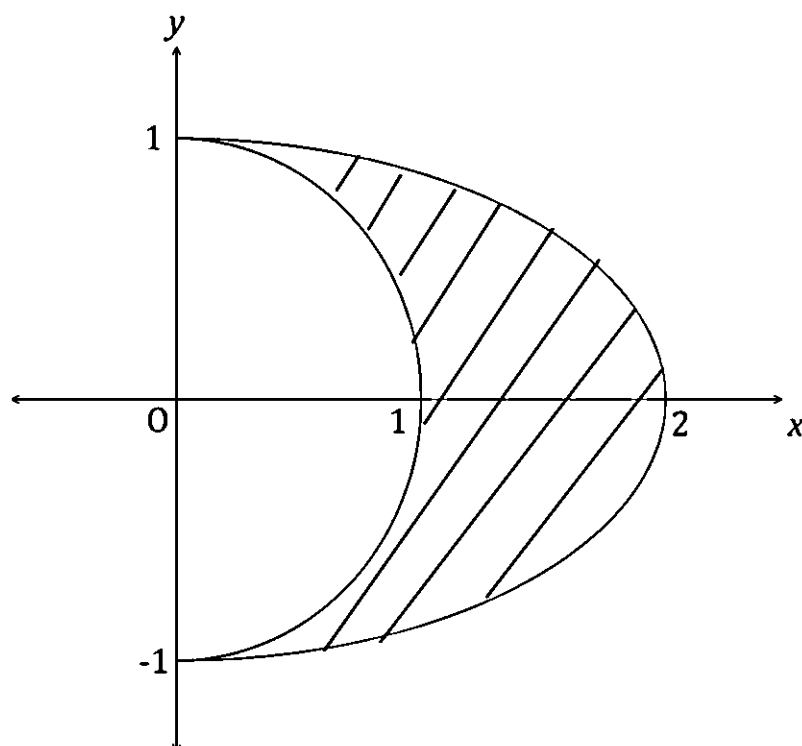
(i) $\cos \theta = \frac{1}{2} \left(z + \frac{1}{z} \right)$ 1

(ii) $\cos n\theta = \frac{1}{2} \left(z^n + \frac{1}{z^n} \right)$ 1

(iii) $\cos^4 \theta = \frac{1}{8} (\cos 4\theta + 4 \cos 2\theta + 3)$ 2

Question 5 continued on page 7

c)



The base of a solid is the shaded region between the semi-circle $x^2 + y^2 = 1$ and the semi-ellipse $\frac{x^2}{4} + y^2 = 1$, $x \geq 0$. Vertical cross-sections taken parallel to the x -axis are rectangles with heights equal to the squares of their base lengths.

- (i) Show that the volume of the solid is given by 2

$$V = \int_{-1}^1 (1 - y^2)^{\frac{3}{2}} dy$$

- (ii) Find the value of V by using the substitution $y = \sin \theta$ and the result of part b). 2

- d) The sequence $\{u_0, u_1, u_2, \dots\}$ is defined by the recurrence relation 3
- $$u_{n+2} - 4u_{n+1} + 4u_n = 0, \quad n = 0, 1, 2, \dots; \text{ and } u_0 = 1, \quad u_1 = 2.$$

Prove by mathematical induction that $u_n = 2^n$ for all integers n .

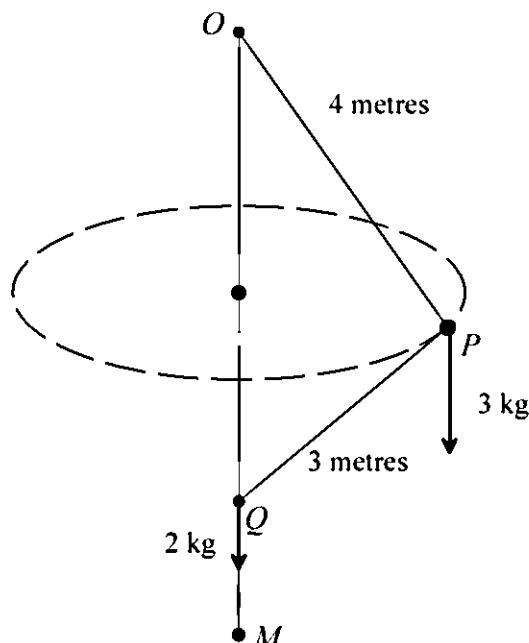
Question 6 (15 marks) Start a NEW PAGE.

MARKS

- a) (i) By using De Moivre's theorem or otherwise, show that 2
 $\cos 5\theta = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta$
- (ii) Solve the equation $32x^5 - 40x^3 + 10x - 1 = 0$. 3
- (iii) Deduce that $\cos \frac{\pi}{15} + \cos \frac{7\pi}{15} + \cos \frac{13\pi}{15} + \cos \frac{19\pi}{15} = -\frac{1}{2}$. 2
- b) (i) By using the result of a) (i) and the identity 2
 $\sin 5\theta = 16 \sin^5 \theta - 20 \sin^3 \theta + 5 \sin \theta$ (DO NOT prove it),
 show that $\tan 5\theta = \frac{t^5 - 10t^3 + 5t}{5t^4 - 10t^2 + 1}$ where $t = \tan \theta$.
- (ii) Deduce that $\tan \frac{\pi}{5} \tan \frac{2\pi}{5} \tan \frac{3\pi}{5} \tan \frac{4\pi}{5} = 5$. 2
- c) α, β are the roots of the equation $x^2 + px + q = 0$, and 2
 $S_n = \alpha^n + \beta^n$ where n is a positive integer, (eg $S_2 = \alpha^2 + \beta^2$).
- (i) Show that $S_{n+2} + p S_{n+1} + q S_n = 0$ 2
- (ii) Hence show that 2
 $(\sqrt{2011} + \sqrt{2010})^4 + (\sqrt{2011} - \sqrt{2010})^4 = 64\,673\,762$

Question 7 (15 marks) Start a NEW PAGE.

a)



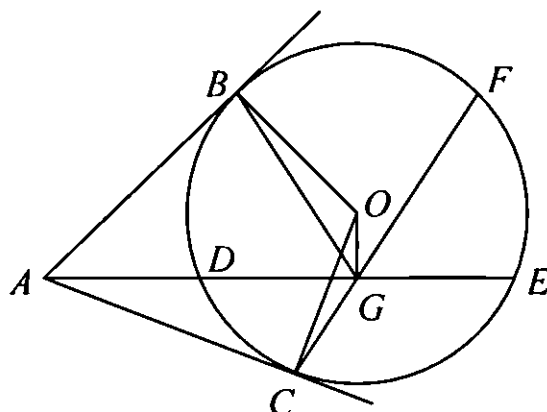
The above sketch shows a smooth vertical rod OM . Light inextensible strings OP and QP are attached to the rod at O and a mass of 3 kg at P . At Q , a 2 kg mass is free to slide on the rod. P is rotating in a horizontal circle about the rod. The distance OQ is 5 metres .

- (i) Calculate, in terms of g , the tension T_1 in PQ and T_2 in OP . 3
- (ii) Hence calculate the angular velocity of P in order to maintain this system. Give your answer correct to one decimal place. Take g as 10 ms^{-2} . 3
- b) A particle of mass $m\text{ kg}$ is projected vertically upwards with an initial velocity 100 ms^{-1} from O . It experiences air resistance during its motion equal to $0.1mv$, where v is its speed in metres per second. Let x be the displacement, in metres, at time t seconds, of the particle measured from O . Take g as 10 ms^{-2} .
- (i) Using a force diagram, explain why $\ddot{x} = -10 - 0.1v$. 1
- (ii) Show that $x = 1000 - 10v - 1000 \times \ln\left(\frac{20}{0.1v + 10}\right)$ 2
- (iii) Find the maximum height reached by the particle. 1
- (iv) Show that $t = 10 \times \ln\left(\frac{20}{0.1v + 10}\right)$ 2
- (v) Hence find the time taken for the particle to reach the maximum height. 1
- (vi) Find the terminal velocity of the particle during its downward journey. 2

Question 8 (15 marks) Start a NEW PAGE.

MARKS

a)



In the diagram, AB and AC are tangents from A to the circle with centre O , meeting the circle at B and C . ADE is a straight line. G is the midpoint of DE . CG produced meets the circle at F .

- (i) Detach the last page from the question booklet and attach it to your answer scripts.
- (ii) Show that $ABOC$ and $AOGC$ are cyclic quadrilaterals. 3
- (iii) Show that $BF \parallel AE$. 2

- b) (i) Show that $\int \ln(1+x) dx = (1+x)\ln(1+x) - x + C$ 1
- (ii) Show that $(n+1)I_n = 2 \ln 2 - \frac{1}{n+1} - nI_{n-1}$, $n = 1, 2, \dots$ 2

$$\text{Where } I_n = \int_0^1 x^n \ln(1+x) dx, \quad n = 0, 1, 2, \dots$$

- (iii) Hence, show that when n is odd, 2

$$(n+1)I_n = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots - \frac{1}{n+1}$$

[You may assume $I_0 = 2 \ln 2 - 1$.]

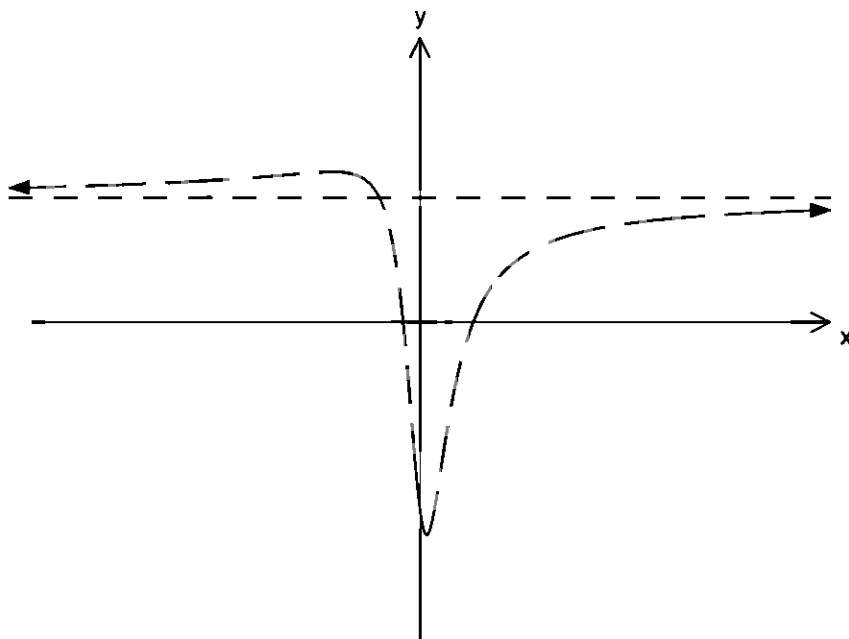
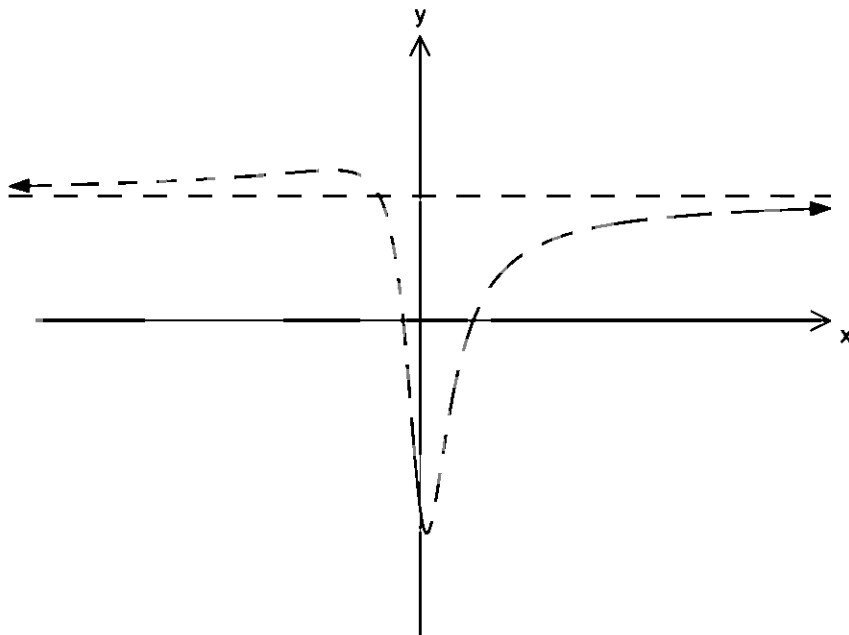
c) Let ω be a complex cube root of unity.

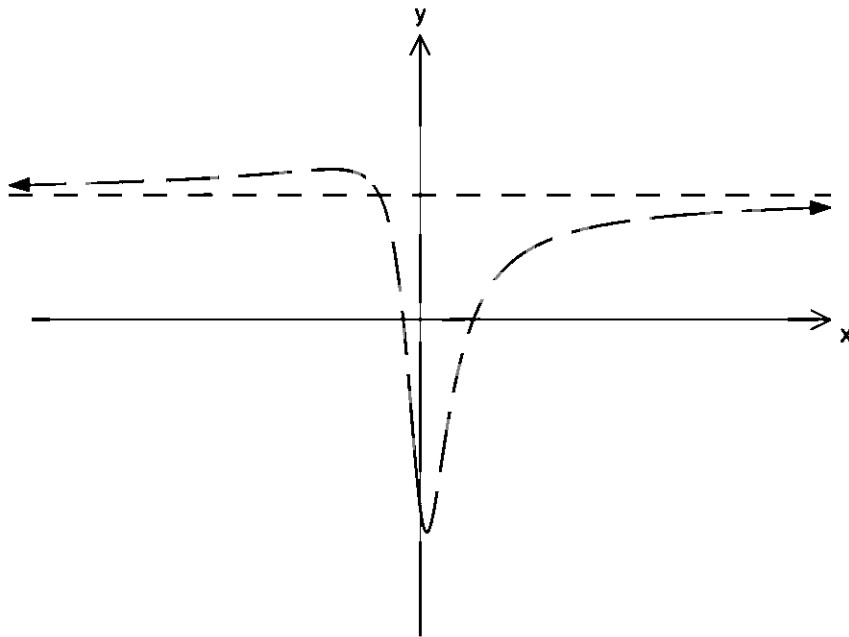
- (i) By using the result $1 + \omega + \omega^2 = 0$, simplify $(1 + \omega)^2$. 1
- (ii) Show that $(1 + \omega)^3 = -1$. 1
- (iii) Use part (ii) to simplify $(1 + \omega)^{3n}$ and hence show that 3

$${}^{3n}C_0 - \frac{1}{2}({}^{3n}C_1 + {}^{3n}C_2) + {}^{3n}C_3 - \frac{1}{2}({}^{3n}C_4 + {}^{3n}C_5) + {}^{3n}C_6 - \dots + {}^{3n}C_{3n} = (-1)^n.$$

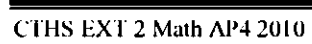
[You may assume $\operatorname{Re}(\omega) = \operatorname{Re}(\omega^2) = -\frac{1}{2}$]

End of Paper





Detach this page and attach it to your answer script.



Solution to C7HS AP4 Ext 2 2010

Question 1

a) Let $u = x^3 - 1$

$\therefore du = 3x^2 dx$

$$\int \frac{x^2}{\sqrt{x^3-1}} dx = \frac{1}{3} \int \frac{du}{\sqrt{u}}$$

$$= \frac{2}{3} \sqrt{u} + C$$

$$= \frac{2}{3} \sqrt{x^3-1} + C \quad \boxed{1}$$

b) $\int \frac{dx}{\sqrt{8x^2+2}} = \frac{1}{2\sqrt{2}} \int \frac{dx}{\sqrt{x^2+\frac{1}{4}}}$ $\boxed{1}$

$$= \frac{1}{2\sqrt{2}} \ln \left[x + \sqrt{x^2 + \frac{1}{4}} \right] + C \quad \boxed{1}$$

or $\frac{1}{2\sqrt{2}} \ln \left[\frac{2x + \sqrt{x^2+1}}{2} \right] + C$

$$= \frac{1}{2\sqrt{2}} \ln [2x + \sqrt{x^2+1}] - \frac{1}{2\sqrt{2}} \ln 2 + C$$

$$= \frac{1}{2\sqrt{2}} \ln [2x + \sqrt{x^2+1}] + C'$$

c) $\int_0^{\frac{1}{2}} \sin^{-1} x dx = [x \sin^{-1} x]_0^{\frac{1}{2}} - \int_0^{\frac{1}{2}} \frac{x}{\sqrt{1-x^2}} dx$ $\boxed{1}$

$$= \frac{\pi}{12} + \frac{1}{2} \int_0^{\frac{1}{2}} \frac{d(1-x^2)}{\sqrt{1-x^2}} \quad \boxed{1}$$

$$= \frac{\pi}{12} + \left[\sqrt{1-x^2} \right]_0^{\frac{1}{2}}$$

$$= \frac{\pi}{12} + \left[\sqrt{\frac{3}{4}} - 1 \right]$$

$$= \frac{\pi}{12} + \frac{\sqrt{3}}{2} - 1 \quad \boxed{1}$$

d) (i) $\frac{x^2-4x+2}{(2x+1)(x^2+4)} = \frac{1}{2x+1} + \frac{Ax+B}{x^2+4}$

$$\therefore (x^2+4) + (Ax+B)(2x+1) = x^2-4x+2 \quad (1)$$

Equating coeff. of x^2

$$2A + 1 = 1$$

$$A = 0 \quad \boxed{1}$$

Put $x=0$ into (1)

$$4 + B = 2$$

$$B = -2 \quad \boxed{1}$$

(ii) $\int_0^2 \frac{x^2-4x+2}{(2x+1)(x^2+4)} dx = \int_0^2 \frac{1}{2x+1} - \frac{2}{x^2+4} dx$

$$= \frac{1}{2} [\ln(2x+1)]_0^2 - \left[\tan^{-1} \frac{x}{2} \right]_0^2 \quad \boxed{1}$$

$$= \frac{1}{2} \ln 5 - \frac{\pi}{4} \quad \boxed{1}$$

Q1 e) Put $t = \tan \frac{\theta}{2}$

$$\therefore d\theta = \frac{2 dt}{1+t^2} \quad [1]$$

$$\left. \begin{array}{l} \text{When } \theta = \frac{\pi}{2}, t = 1 \\ \theta = 0, t = 0 \end{array} \right\} [1]$$

$$\therefore \int_0^{\frac{\pi}{2}} \frac{d\theta}{1 + \sin\theta + \cos 2\theta}$$

$$= \int_0^1 \frac{2 dt}{(1+t^2)\left(1 + \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2}\right)}$$

$$= \int_0^1 \frac{2 dt}{1+t^2 + 2t + 1 - t^2} \quad [1]$$

$$= \int_0^1 \frac{2 dt}{2(1+t)}$$

$$= \int_0^1 \frac{dt}{1+t}$$

$$= [\ln(1+t)]_0^1 \quad [1]$$

$$= \ln 2$$

Question 2

$$\begin{aligned} \text{a) (i) } z - \bar{w} &= (5-12i) + (3+4i) \\ &= 5-12i+3+4i \\ &= 8-16i \end{aligned} \quad [1]$$

$$\begin{aligned} \text{(ii) } \frac{z}{w} &= \frac{5-12i}{3+4i} \\ &= \frac{5-12i}{3+4i} \times \frac{3-4i}{3-4i} \end{aligned} \quad [1]$$

$$= \frac{15-56i-48}{9+16}$$

$$= \frac{-33-56i}{25}$$

$$= -\frac{33}{25} - \frac{56}{25}i \quad [1]$$

$$\begin{aligned} \text{b) (i) } \tan \frac{\pi}{12} &= \tan \left(\frac{\pi}{3} - \frac{\pi}{4} \right) \\ &= \frac{\tan \frac{\pi}{3} - \tan \frac{\pi}{4}}{1 + \tan \frac{\pi}{3} \tan \frac{\pi}{4}} \quad [1] \\ &= \frac{\sqrt{3} - 1}{1 + \sqrt{3}} \end{aligned}$$

$$\begin{aligned} \text{(ii) } z &= (\sqrt{3}+1) + (\sqrt{3}-1)i \\ |z| &= \sqrt{(\sqrt{3}+1)^2 + (\sqrt{3}-1)^2} \\ &= \sqrt{3+2\sqrt{3}+1+3-2\sqrt{3}+1} \\ &= \sqrt{8} \end{aligned} \quad [1]$$

$$\arg Z = \tan^{-1} \frac{\sqrt{3}-1}{\sqrt{3}+1}$$

$$= \frac{\pi}{12} \text{ from b) (i)}$$

[1]

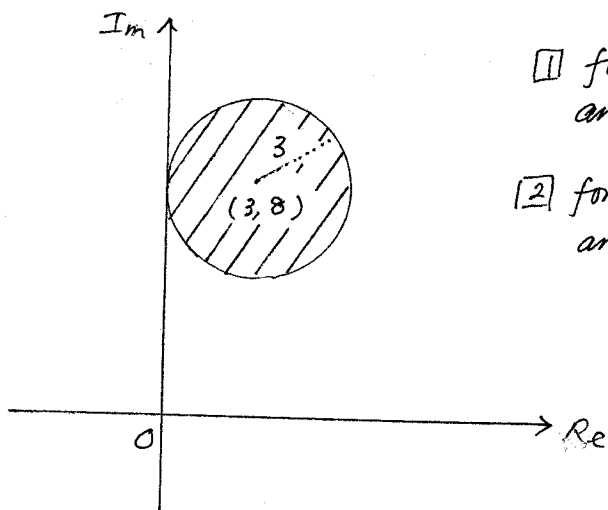
$$\therefore (\sqrt{3}+1) + (\sqrt{3}-1)i = \sqrt{8} \operatorname{cis} \frac{\pi}{12}$$

$$(iii) Z^n = \sqrt{8}^n \operatorname{cis} \frac{n\pi}{12}$$

$$\operatorname{Im}(Z^n) = \sqrt{8}^n \sin \frac{n\pi}{12}$$

The least positive integer for which $\sin \frac{n\pi}{12} = 0$ is 12 [1]

c) (i)



[1] for correct centre and radius

[2] for correct region and closed boundary.

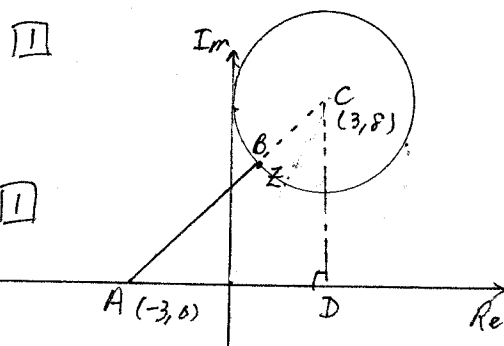
(ii) The min. value of $|Z+3|$

$$= AB$$

$$= AC - BC$$

$$= \sqrt{6^2 + 8^2} - 3$$

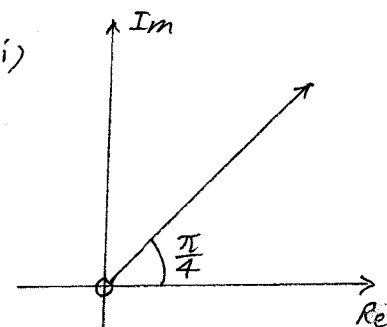
$$= 7$$



[1]

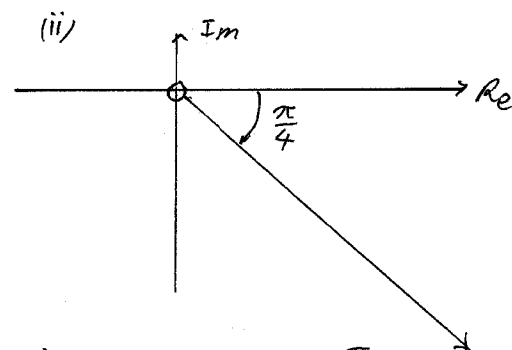
[1]

d) (i)



[1]

(ii)



[1]

e) (i) Since $\angle AOB = \frac{\pi}{2}$

$$\therefore \vec{OB} = 2\vec{OA} \operatorname{cis} \frac{\pi}{2}$$

$$= 2i\vec{OA}$$

$$= 2i(x+iy)$$

$$= 2ix - 2y$$

$$= -2y + 2ix$$

$$(ii) \vec{OC} = \vec{OA} + \vec{OB}$$

$$= (x+iy) + (-2y+2ix)$$

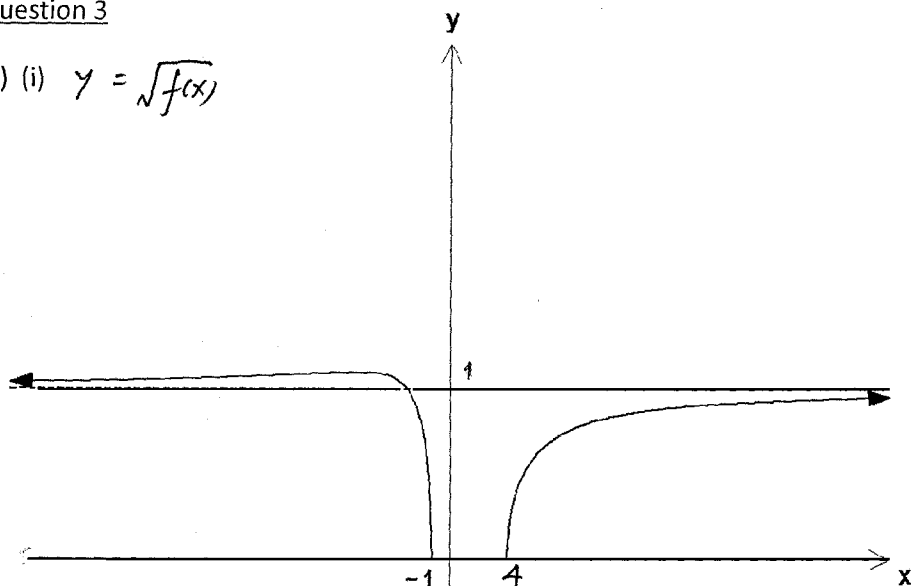
$$= (x-2y) + (2x+y)i$$

[1]

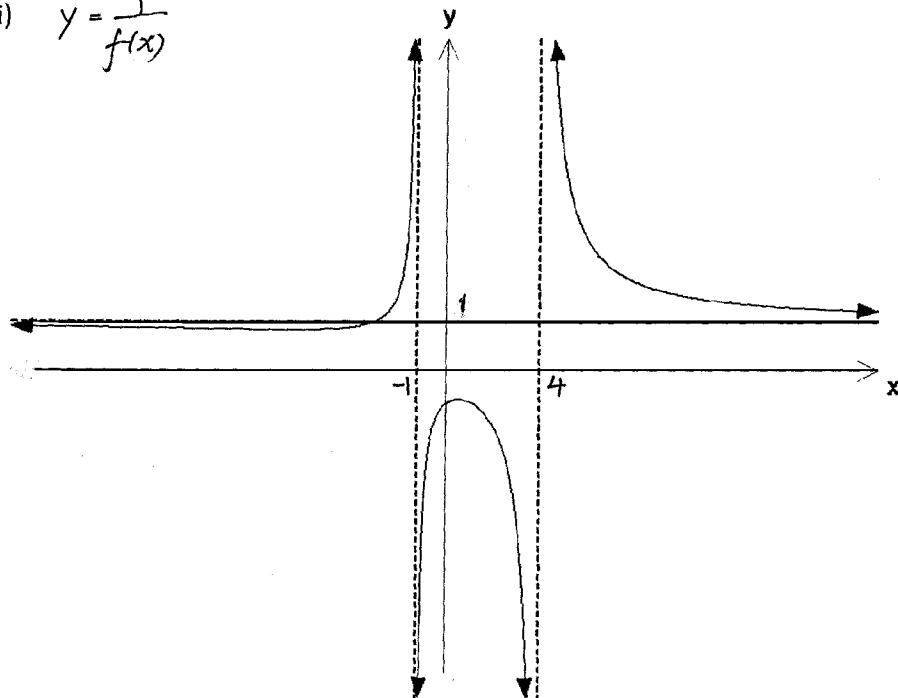
[1]

Question 3

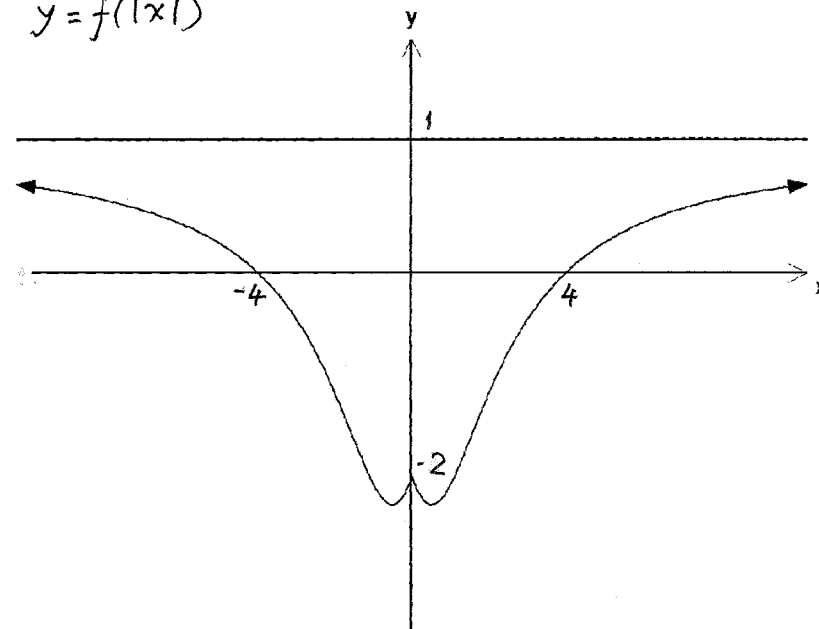
a) (i) $y = \sqrt{f(x)}$



(ii) $y = \frac{1}{f(x)}$



(iii) $y = f(|x|)$



b) (i) The only possible integer roots are 1 or -1.

But $P(1) = -1 \neq 0$

$P(-1) = 10 \neq 0$

$\therefore P(x) = 0$ has no integer roots.

(ii) $P(0) = 1$

$P(0)$ and $P(1)$ are opposite in sign, hence there must be a real root.

(iii) $\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = (\sum \alpha)^2 - 2 \sum \alpha\beta$
 $= 2^2 - 2(3)$
 $= -2$

(iv) Since $\sum \alpha^2 < 0$

$\therefore$ Not all roots are real.

But there is a real root between 0 & 1 by (i) and complex roots occur in pairs, there can only be 2 complex roots.

Hence there must be 2 real roots.

Q 3(b)

$$(v) \frac{1}{2\alpha\beta\gamma} = \frac{\delta}{2\alpha\beta\gamma\delta}$$

$$= \frac{\delta}{2} \quad \text{since } \alpha\beta\gamma\delta = 1$$

$$\therefore \text{ let } y = \frac{x}{2}$$

$$x = 2y$$

$$P(x) = x^4 - 2x^3 + 3x^2 - 4x + 1 = 0$$

$$(2y)^4 - 2(2y)^3 + 3(2y)^2 - 4(2y) + 1 = 0 \quad [1]$$

$$16y^4 - 16y^3 + 12y^2 - 8y + 1 = 0$$

$\therefore$ The required equation is

$$16x^4 - 16x^3 + 12x^2 - 8x + 1 = 0 \quad [1]$$

$$(vi) \frac{1}{2\alpha\beta\gamma} + \frac{1}{2\alpha\beta\delta} + \frac{1}{2\alpha\gamma\delta} + \frac{1}{2\beta\gamma\delta}$$

= sum of roots of the new equation

$$= 1 \quad [1]$$

Question 4

a) (i) Foci are $(2, -1)$ and $(10, -1)$

$\therefore$ Centre is $(6, -1)$

$\therefore$ The complex number corresponding to the centre of the ellipse is $6 - i$ [1]

(ii)

$$2a = 12$$

$$a = 6$$

$$ae = 4$$

$$6e = 4$$

$$e = \frac{2}{3}$$

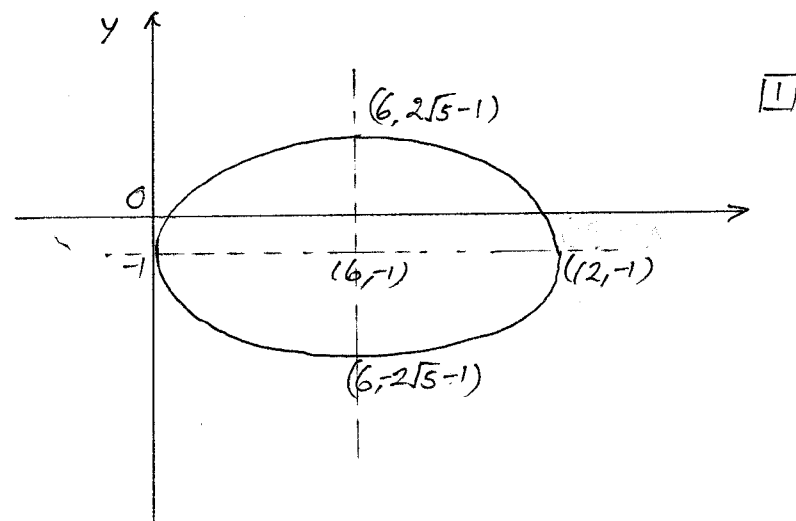
$$b = a\sqrt{1-e^2}$$

$$= 6\sqrt{1-\frac{4}{9}}$$

$$= 2\sqrt{5}$$

$\therefore$ major axis = 12 [1]

minor axis = $4\sqrt{5}$ [1]



b) (i) The tangent at P is $\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$ [1]

(ii) S is $(ae, 0)$

$$\therefore ST = \frac{\left| \frac{ae \cos \theta}{a} - 1 \right|}{\sqrt{\left(\frac{\cos \theta}{a} \right)^2 + \left(\frac{\sin \theta}{b} \right)^2}} \quad [1]$$

Q 4 (b)(ii) cont'd

$$ST = \frac{ab |e \cos \theta - 1|}{ab \sqrt{\frac{\cos^2 \theta}{a^2} + \frac{\sin^2 \theta}{b^2}}}$$

$$= \frac{ab |e \cos \theta - 1|}{\sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}} \quad [1]$$

(iii) Similarly $S'T' = \frac{ab |-e \cos \theta - 1|}{\sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}}$

$$= \frac{ab |e \cos \theta + 1|}{\sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}}$$

$$\therefore ST \times S'T' = \frac{ab |e \cos \theta - 1|}{\sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}} \times \frac{ab |e \cos \theta + 1|}{\sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}}$$

$$= \frac{a^2 b^2 (1 - e \cos \theta)(1 + e \cos \theta)}{a^2 \sin^2 \theta + b^2 \cos^2 \theta}$$

$$= \frac{a^2 b^2 (1 - e^2 \cos^2 \theta)}{a^2 \sin^2 \theta + b^2 \cos^2 \theta} \quad [1]$$

$$= \frac{b^2 (a^2 - a^2 e^2 \cos^2 \theta)}{a^2 \sin^2 \theta + b^2 \cos^2 \theta}$$

$$= \frac{b^2 [a^2 - (a^2 - b^2) \cos^2 \theta]}{a^2 \sin^2 \theta + b^2 \cos^2 \theta}$$

$$= \frac{b^2 [(a^2 - a^2 \cos^2 \theta) + b^2 \cos^2 \theta]}{a^2 \sin^2 \theta + b^2 \cos^2 \theta} \quad [1]$$

$$= \frac{b^2 (a^2 \sin^2 \theta + b^2 \cos^2 \theta)}{a^2 \sin^2 \theta + b^2 \cos^2 \theta}$$

$$= b^2$$

(c) (i) $xy = c^2$

$$y = \frac{c^2}{x}$$

$$\therefore y' = -\frac{c^2}{x^2}$$

$\therefore$ gradient of tangent at A is $m_1 = -\frac{c^2}{c^2 a^2}$

$$= -\frac{1}{a^2}$$

Gradient of normal at B is $m_2 = a^2$ [1]

$\therefore$ Equation of normal at A is

$$y - \frac{c}{a} = a^2 (x - ca)$$

$$= a^2 x - ca^3$$
[1]

ie $y = a^2 x - ca^3 + \frac{c}{a}$

$$y = a^2 x + \frac{c}{a} (1 - a^4) \quad (1)$$

(ii) B lies on the normal at A

$\therefore (cb, \frac{c}{b})$ satisfies (1)

$$\frac{c}{b} = a^2 cb + \frac{c}{a} (1 - a^4)$$

$$ac = a^3 cb^2 + bc(1 - a^4)$$

$$a = a^3 b^2 + b - a^4 b$$

$$a - b = a^3 b (b - a)$$
[1]

$$1 = -a^3 b$$

$$\therefore b = -\frac{1}{a^3}$$

[alternatively]

when $y = a^2x + \frac{c}{a}(1-a^4)$ meets $xy = c^2$

$$\frac{c^2}{x} = a^2x + \frac{c}{a}(1-a^4)$$

$$ac^2 = a^3x^2 + c(1-a^4)x$$

$$a^3x^2 + c(1-a^4)x - ac^2 = 0$$

The x-coordinates of A and B are solution of this equation

$$\text{Product of roots} = -\frac{ac^2}{a^3}$$

$$= -\frac{c^2}{a^2}$$

$$\therefore (ca)(cb) = -\frac{c^2}{a^2}$$

$$ab = -\frac{1}{a^2}$$

$$b = -\frac{1}{a^3}$$

$$\text{iii) At M } x = \frac{c}{2}(a+b)$$

$$y = \frac{c}{2}\left(\frac{1}{a} + \frac{1}{b}\right)$$

$$y = \frac{c}{2} \frac{a+b}{ab}$$

$$\frac{x}{y} = ab$$

$$\text{but } b = -\frac{1}{a^3}$$

$$\text{Put into (3) } \frac{x}{y} = -\frac{1}{a^2}$$

$$\text{or } a^2 = -\frac{y}{x}$$

$$\begin{aligned} \text{(iv) From (2)} \quad y &= \frac{c}{2}\left(\frac{1}{a} + \frac{1}{b}\right) \\ &= \frac{c}{2}\left(\frac{1}{a} - a^3\right) \end{aligned}$$

$$= \frac{c}{2} \left(\frac{1-a^4}{a} \right)$$

$$y^2 = \frac{c^2}{4} \frac{(1-a^4)^2}{a^2} \quad (5)$$

Put (4) into (5)

$$y^2 = \frac{c^2}{4} \frac{\left[1 - \left(-\frac{y}{x}\right)^2\right]^2}{\left(-\frac{y}{x}\right)}$$

$$= \frac{c^2}{4} \frac{\left[1 - \frac{y^2}{x^2}\right]^2}{-\left(\frac{y}{x}\right)}$$

$$= -\frac{c^2}{4} \frac{(x^2 - y^2)^2}{x^4} \cdot \frac{x}{y}$$

$$= -\frac{c^2(x^2 - y^2)^2}{4x^3y}$$

$$\text{i.e. } 4x^3y^3 + c^2(x^2 - y^2)^2 = 0$$

Question 5

$$\begin{aligned} \text{a) } h &= 1 - x(2x-1)^2 \\ r &= x \end{aligned} \quad \left. \vphantom{\begin{aligned} h &= 1 - x(2x-1)^2 \\ r &= x \end{aligned}} \right\} \quad \square$$

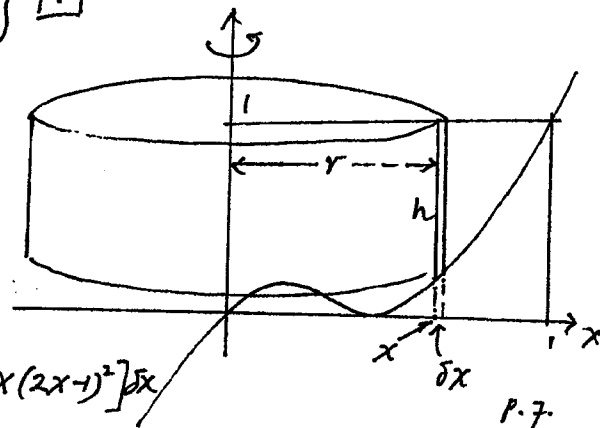
Vol of the cylindrical shell

$$\delta V = 2\pi r h \delta x$$

$$= 2\pi x [1 - x(2x-1)^2] \delta x$$

$\therefore$ Vol of solid

$$V = \lim_{\delta x \rightarrow 0} \sum_{x=0}^1 2\pi x [1 - x(2x-1)^2] \delta x$$



Q 5 (cont'd)

$$V = \int_0^1 2\pi x [1 - x(2x-1)^2] dx \quad [1]$$

$$= 2\pi \int_0^1 x - x^2(2x-1)^2 dx$$

$$= \pi [x^2]_0^1 - 2\pi \int_0^1 x^2(4x^2 - 4x + 1) dx$$

$$= \pi - 2\pi \int_0^1 (4x^4 - 4x^3 + x^2) dx \quad [1]$$

$$= \pi - 2\pi \left[\frac{4x^5}{5} - x^4 + \frac{x^3}{3} \right]_0^1$$

$$= \pi - 2\pi \left(\frac{4}{5} - 1 + \frac{1}{3} \right)$$

$$= \pi - \frac{4\pi}{15}$$

$$= \frac{11\pi}{15} \text{ unit}^3 \quad [1]$$

b) (i) $z = \cos\theta + i\sin\theta$ (1)

$$\frac{1}{z} = \cos\theta - i\sin\theta \quad (2)$$

$$(1)+(2) \quad z + \frac{1}{z} = 2\cos\theta \quad [1]$$

$$\therefore 2\cos\theta = z + \frac{1}{z}$$

$$\cos\theta = \frac{1}{2} \left(z + \frac{1}{z} \right)$$

$$(ii) \quad z^n = \cos n\theta + i\sin n\theta \quad (3)$$

$$\frac{1}{z^n} = \cos n\theta - i\sin n\theta \quad (4)$$

$$(3)+(4) \quad z^n + \frac{1}{z^n} = 2\cos n\theta \quad [1]$$

$$\text{i.e. } \cos n\theta = \frac{1}{2} \left(z^n + \frac{1}{z^n} \right)$$

$$(iii) \quad (2\cos\theta)^4 = \left(z + \frac{1}{z} \right)^4$$

$$16\cos^4\theta = z^4 + 4z^2 + 6 + \frac{4}{z^2} + \frac{1}{z^4} \quad [1]$$

$$= \left(z^4 + \frac{1}{z^4} \right) + 4 \left(z^2 + \frac{1}{z^2} \right) + 6$$

$$= 2\cos 4\theta + 8\cos 2\theta + 6$$

$$\therefore \cos^4\theta = \frac{1}{8} (\cos 4\theta + 4\cos 2\theta + 3) \quad [1]$$

(c) For the circle $x^2 + y^2 = 1$

$$(i) \quad x_1 = \sqrt{1-y^2}$$

For the ellipse $\frac{x}{4} + y^2 = 1$

$$x_2 = 2\sqrt{1-y^2}$$

$$\therefore l = x_2 - x_1$$

$$= 2\sqrt{1-y^2} - \sqrt{1-y^2}$$

$$= \sqrt{1-y^2}$$

Height of the rectangle = l^2

$$\text{i.e. } h = 1-y^2 \quad [1]$$

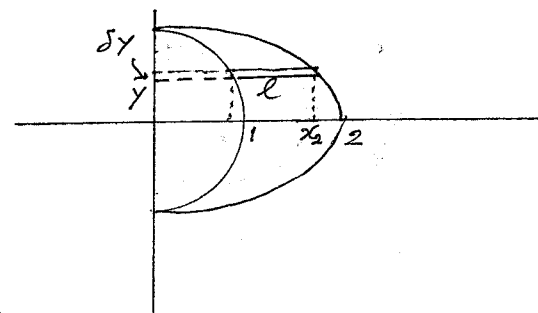
$\therefore$ Area of cross-section = hl

$$= (1-y^2)\sqrt{1-y^2}$$

$$= (1-y^2)^{3/2}$$

$\therefore$ Volume of the vertical slice is

$$\delta V = (1-y^2)^{3/2} \delta y \quad [1]$$



Q 5 (cont'd)

Volume of the solid

$$\begin{aligned} V &= \lim_{\delta y \rightarrow 0} \sum \delta V \\ &= \lim_{\delta y \rightarrow 0} \sum_{y=-1}^1 (1-y^2)^{3/2} \delta y \\ &= \int_{-1}^1 (1-y^2)^{3/2} dy \end{aligned}$$

Qiv Let $y = \sin \theta$

$$dy = \cos \theta d\theta$$

$$\text{when } y=1 \quad \theta = \frac{\pi}{2}$$

$$\text{when } y=0 \quad \theta = 0$$

$$\therefore V = 2 \int_0^1 (1-y^2)^{3/2} dy \quad \because (1-y^2)^{3/2} \text{ is an even function of } y.$$

$$= 2 \int_0^{\pi/2} (1-\sin^2 \theta)^{3/2} \cos \theta d\theta$$

$$= 2 \int_0^{\pi/2} \cos^3 \theta \cdot \cos \theta d\theta$$

$$= 2 \int_0^{\pi/2} \cos^4 \theta d\theta$$

$$= \frac{2}{8} \int_0^{\pi} (\cos 4\theta + 4 \cos 2\theta + 3) d\theta$$

$$= \frac{1}{4} \left[\frac{1}{4} \sin 4\theta + 2 \sin 2\theta + 3\theta \right]_0^{\pi}$$

$$= \frac{1}{4} \left[0 + 0 + \frac{3\pi}{2} \right]$$

$$= \frac{3\pi}{8} \text{ unit}^3$$

$$5d) u_2 = 4u_1 - 4u_0$$

$$= 4(2) - 4(1)$$

$$= 4$$

$$= 2^2$$

$\therefore$ It is true for $n=2$

$$u_3 = 4u_2 - 4u_1$$

$$= 4(4) - 4(2)$$

$$= 8$$

$$= 2^3$$

$\therefore$ True for $n=3$.

Assume it is true for $n=k$ and $k+1$

$$\text{ie } u_k = 2^k, \quad u_{k+1} = 2^{k+1}$$

$$\text{then } u_{k+2} = 4u_{k+1} - 4u_k$$

$$= 4(2^{k+1}) - 4(2^k) \quad \text{by assumption}$$

$$= 4(2^{k+1} - 2^k)$$

$$= 4(2 \cdot 2^k - 2^k)$$

$$= 4(2^k)$$

$$= 2^{k+2}$$

ie true for $n=k+2$ if true for $n=k$ and $n=k+1$,

since it is proved true for $n=2$ and 3 , hence it is true for $n=4, 5, 6, \dots$ ie true for all positive integers.

Question 6

$$\begin{aligned}
 \text{a) (i)} \quad \cos 5\theta + i \sin 5\theta &= (\cos \theta + i \sin \theta)^5 \\
 &= \cos^5 \theta + 5(\cos^4 \theta)(i \sin \theta) + 10(\cos^3 \theta)(i \sin \theta)^2 + 10(\cos^2 \theta)(i \sin \theta)^3 \\
 &\quad + 5(\cos \theta)(i \sin \theta)^4 + (i \sin \theta)^5 \quad [1] \\
 &= \cos^5 \theta + 5i \cos^4 \theta \sin \theta - 10 \cos^3 \theta \sin^2 \theta - 10i \cos^2 \theta \sin^3 \theta + 5 \cos \theta \sin^4 \theta \\
 &\quad + i \sin^5 \theta
 \end{aligned}$$

Equating real parts

$$\begin{aligned}
 \cos 5\theta &= \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta \quad [1] \\
 &= \cos^5 \theta - 10 \cos^3 \theta (1 - \cos^2 \theta) + 5 \cos \theta (1 - \cos^2 \theta)^2 \\
 &= \cos^5 \theta - 10 \cos^3 \theta + 10 \cos^5 \theta + 5 \cos \theta - 10 \cos^3 \theta + 5 \cos^5 \theta \\
 &= 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta
 \end{aligned}$$

(ii) Let $x = \cos \theta$

$$32x^5 - 40x^3 + 10x - 1 = 0 \quad (1)$$

$$16x^5 - 20x^3 + 5x = \frac{1}{2}$$

$$16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta = \frac{1}{2}$$

$$\cos 5\theta = \frac{1}{2} \quad [1]$$

$$5\theta = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}, \frac{13\pi}{3}$$

$$\theta = \frac{\pi}{15}, \frac{\pi}{3}, \frac{7\pi}{15}, \frac{11\pi}{15}, \frac{13\pi}{15} \quad [1]$$

$$\therefore \text{Roots are } \cos \frac{\pi}{15}, \cos \frac{\pi}{3}, \cos \frac{7\pi}{15}, \cos \frac{11\pi}{15}, \cos \frac{13\pi}{15} \quad [1]$$

(iii) Sum of roots of (1)

$$\cos \frac{\pi}{15} + \cos \frac{\pi}{3} + \cos \frac{7\pi}{15} + \cos \frac{11\pi}{15} + \cos \frac{13\pi}{15} = 0 \quad [1]$$

$$\cos \frac{\pi}{15} + \frac{1}{2} + \cos \frac{7\pi}{15} + \cos \left(2\pi - \frac{19\pi}{15}\right) + \cos \frac{13\pi}{15} = 0 \quad [1]$$

$$\cos \frac{\pi}{15} + \cos \frac{7\pi}{15} + \cos \frac{13\pi}{15} + \cos \frac{19\pi}{15} = -\frac{1}{2}$$

b)

$$\tan 5\theta = \frac{\sin 5\theta}{\cos 5\theta}$$

$$\begin{aligned}
 &= \frac{16 \sin^5 \theta - 20 \sin^3 \theta + 5 \sin \theta}{16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta} \\
 &= \frac{(16 \sin^5 \theta - 20 \sin^3 \theta + 5 \sin \theta) \times \frac{1}{\cos^5 \theta}}{(16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta) \times \frac{1}{\cos^5 \theta}} \quad [1] \\
 &= \frac{16 \tan^5 \theta - 20 \tan^3 \theta \sec^2 \theta + 5 \tan \theta \cdot \sec^4 \theta}{16 - 20 \sec^2 \theta + 5 \sec^4 \theta} \\
 &= \frac{16 \tan^5 \theta - 20 \tan^3 \theta (1 + \tan^2 \theta) + 5 \tan \theta (1 + \tan^2 \theta)^2}{16 - 20(1 + \tan^2 \theta) + 5(1 + \tan^2 \theta)^2} \quad [1] \\
 &= \frac{16 \tan^5 \theta - 20 \tan^3 \theta - 20 \tan^5 \theta + 5 \tan \theta + 10 \tan^3 \theta + 5 \tan \theta}{16 - 20 - 20 \tan^2 \theta + 5 + 10 \tan^2 \theta + 5 \tan^4 \theta} \\
 &= \frac{\tan^5 \theta - 10 \tan^3 \theta + 5 \tan \theta}{5 \tan^4 \theta - 10 \tan^2 \theta + 1}
 \end{aligned}$$

Q6 (cont'd)

b) (ii) Consider $\tan 5\theta = 0$

$$\therefore 5\theta = \pi, 2\pi, 3\pi, 4\pi, 5\pi, \dots$$

$$\theta = \frac{\pi}{5}, \frac{2\pi}{5}, \frac{3\pi}{5}, \frac{4\pi}{5}, \pi, \dots$$

but $\tan 5\theta = \frac{t^5 - 10t^3 + 5t}{5t^4 - 10t^2 + 1}$ where $t = \tan \theta$

$$\therefore \text{The solution to } \frac{t^5 - 10t^3 + 5t}{5t^4 - 10t^2 + 1} = 0$$

$$\text{ie } t^5 - 10t^3 + 5t = 0$$

$$\text{are } t = \tan \frac{\pi}{5}, \tan \frac{2\pi}{5}, \tan \frac{3\pi}{5}, \tan \frac{4\pi}{5}, \tan \pi \quad [1]$$

$$t = \tan \pi \text{ corresponding to } t = 0$$

$$\therefore \text{Roots of } t^4 - 10t^2 + 5 = 0$$

$$\text{are } \tan \frac{\pi}{5}, \tan \frac{2\pi}{5}, \tan \frac{3\pi}{5}, \tan \frac{4\pi}{5} \quad [1]$$

Product of roots:

$$\tan \frac{\pi}{5} \tan \frac{2\pi}{5} \tan \frac{3\pi}{5} \tan \frac{4\pi}{5} = 5$$

c) (i) Since α, β are roots of $x^2 + px + q = 0$

$$\therefore \alpha^2 + p\alpha + q = 0 \quad (1)$$

$$\beta^2 + p\beta + q = 0 \quad (2)$$

$$\alpha^{n+2} + p\alpha^{n+1} + q\alpha^n = 0 \quad (3)$$

$$\beta^{n+2} + p\beta^{n+1} + q\beta^n = 0 \quad (4)$$

$$(1) \cdot \alpha^n$$

$$(2) \cdot \beta^n$$

} [1]

$$(3) + (4) \quad (\alpha^{n+2} + \beta^{n+2}) + p(\alpha^{n+1} + \beta^{n+1}) + q(\alpha^n + \beta^n) = 0 \quad [1]$$

$$S_{n+2} + pS_{n+1} + qS_n = 0$$

$$(ii) \text{ Let } \alpha = \sqrt{2011} + \sqrt{2010}, \beta = \sqrt{2011} - \sqrt{2010}$$

$$\therefore \alpha + \beta = 2\sqrt{2011}, \quad \alpha\beta = 2011 - 2010 = 1$$

$\therefore \alpha$ and β are roots of the equation

$$x^2 - 2\sqrt{2011}x + 1 = 0 \quad [1]$$

$$\begin{aligned} S_2 &= \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta \\ &= 4(2011) - 2 \\ &= 8042 \end{aligned}$$

$$S_3 - 2\sqrt{2011}S_2 + S_1 = 0$$

$$\begin{aligned} \therefore S_3 &= 2\sqrt{2011}S_2 - S_1 \\ &= 16084\sqrt{2011} - 2\sqrt{2011} \\ &= 16082\sqrt{2011} \end{aligned}$$

$$S_4 - 2\sqrt{2011}S_3 + S_2 = 0 \quad [1]$$

$$\begin{aligned} S_4 &= 2\sqrt{2011}S_3 - S_2 \\ &= 2\sqrt{2011}(16082\sqrt{2011}) - 8042 \\ &= 64673762. \end{aligned}$$

Question 7

a) (i) $OP = 4$, $PQ = 3$, $OQ = 5$

$\therefore \angle OPQ = 90^\circ$

$\cos \theta = \frac{4}{5}$, $\sin \theta = \frac{3}{5}$

Resolving forces vertically at Q:

$T_1 \sin \theta = 2g$

$\frac{3T_1}{5} = 2g$

$\therefore T_1 = \frac{10g}{3}$ [1]

Resolving forces vertically at P

$T_2 \cos \theta = T_1 \sin \theta + 3g$ [1]

$\frac{4T_2}{5} = \frac{3T_1}{5} + 3g$

$= \frac{3}{5} \left(\frac{10g}{3} \right) + 3g$

$= 5g$

$\therefore T_2 = \frac{25g}{4}$ [1]

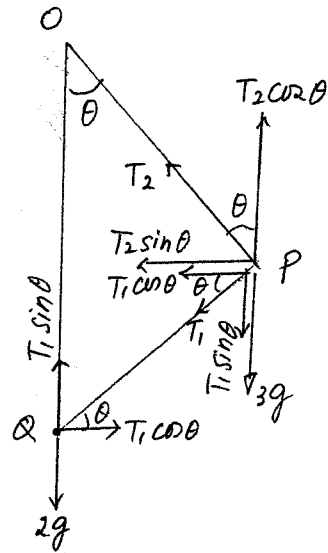
(ii) Radius of the horizontal circle

$r = 4 \sin \theta$

$= \frac{12}{5}$ [1]

Resolving forces horizontally at P

$T_1 \cos \theta + T_2 \sin \theta = 3\omega^2 r$ [1]



$\frac{10g}{3} \cdot \frac{4}{5} + \frac{25g}{4} \cdot \frac{3}{5} = 3 \cdot \frac{12}{5} \omega^2$

$\therefore \omega^2 = \frac{385g}{432}$

$\therefore \omega = \sqrt{\frac{385g}{432}}$

$= 2.985$

$= 3.0 \text{ rad s}^{-1} \text{ (1 d.p.)}$ [1]

b) (i) Take upward direction as positive

$m\ddot{x} = -mg - 0.1mv$

$\therefore \ddot{x} = -g - 0.1v$ [1]

(ii) $v \frac{dv}{dx} = -(g + 0.1v)$

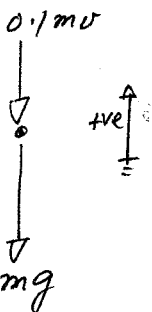
$\therefore \int_{100}^v \frac{v dv}{g + 0.1v} = - \int_0^x dx$ [1]

$\int_{100}^v \left(10 - \frac{10g}{g + 0.1v} \right) dv = -[x]_0^x$

$[10v - 100g \ln(g + 0.1v)]_{100}^v = -x$ [1]

$10v - 100g \ln(g + 0.1v) - 1000 + 100g \ln(g + 10) = -x$

$\therefore x = 1000 - 10v - 100g \ln \left(\frac{g + 10}{g + 0.1v} \right)$



Q7 (cont'd)

$$x = 1000 - 10v - 1000 \ln\left(\frac{20}{10+0.1v}\right)$$

$$g = 10$$

(iii) At max height, $v = 0$

$$x_{\max} = 1000 - 1000 \ln\left(\frac{20}{10}\right) \\ = 1000 - 1000 \ln 2$$

[1]

(iv) From $\ddot{x} = -(g + 0.1v)$

$$\frac{dv}{dt} = -(10 + 0.1v)$$

$$\int_{100}^v \frac{dv}{10 + 0.1v} = - \int_0^t dt$$

[1]

$$10 \ln[10 + 0.1v]_{100}^v = -[t]_0^t$$

$$10 \ln\left(\frac{10 + 0.1v}{20}\right) = -t$$

$$\therefore t = 10 \ln\left(\frac{20}{10 + 0.1v}\right)$$

[1]

(v) when $v = 0$, $t = 10 \ln \frac{20}{10}$
 $= 10 \ln 2$

[1]

(vi) During downward journey

$$m\ddot{x} = mg - 0.1mv$$

$$\ddot{x} = g - 0.1v$$

$$= 10 - 0.1v$$

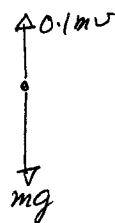
when $v = v_T$, $\ddot{x} = 0$

$$\therefore 0 = 10 - 0.1v_T$$

$$v_T = 100 \text{ m s}^{-1}$$

[1]

[1]



Question 8

a) (ii) $\angle ABO = \angle ACO = 90^\circ$

(radius $\perp$ tangent at pt of contact)

$$\angle ABO + \angle ACO = 2 \text{ rt } \angle\text{'s}$$

$\therefore ABOC$ is a cyclic quad.

(opp $\angle$'s of cyclic quad are supplementary)

In $\triangle ODG$, $\triangle OEG$

$$OD = OE \text{ (radii)}$$

$$DG = EG \text{ (given)}$$

$$OG = OG \text{ (common)}$$

$$\therefore \triangle DGO \equiv \triangle EGO \text{ (SSS)}$$

$$\therefore \angle DGO = \angle EGO \text{ (corr. } \angle\text{'s of congruent } \triangle\text{'s)}$$

$$\text{but } \angle DGO + \angle EGO = 180^\circ \text{ (}\angle\text{'s on a str. line)}$$

$$\therefore \angle DGO = 90^\circ$$

$$= \angle OCA$$

$\therefore AOGC$ is a cyclic quad. ($\angle$'s in same segment are equal)

(iii) $\angle FGE = \angle AGC$ (vert. opp $\angle$'s)

$$\angle AGC = \angle AOC$$

($\angle$'s in same segment, $AOGC$ is a cyclic quad)

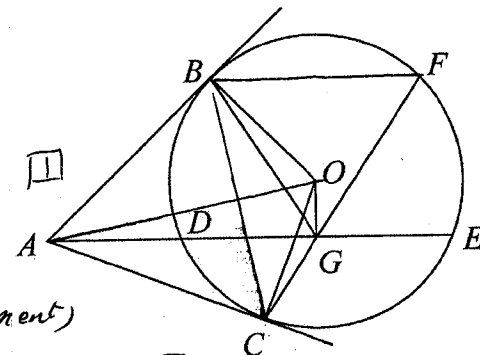
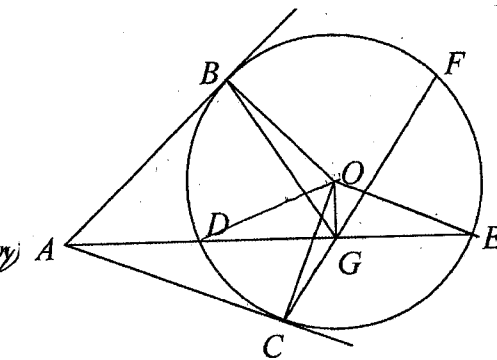
$$\angle AOC = \angle ABC$$

($\angle$'s in same segment, $ABOC$ is cyclic quad)

$$\angle ABC = \angle BFC \text{ (}\angle\text{'s in alt segment)}$$

$$\therefore \angle FGE = \angle BFG$$

$$BF \parallel AE \text{ (alt } \angle\text{'s on } \parallel \text{ lines are equal)}$$



$$\begin{aligned} \text{b) ii) } \frac{d}{dx} [(1+x) \ln(1+x) - x] \\ = \ln(1+x) + \frac{1+x}{1+x} - 1 \\ = \ln(1+x) \end{aligned}$$

$$\begin{aligned} \text{b) ii) } \frac{d}{dx} [(1+x) \ln(1+x) - x] \\ = \ln(1+x) + \frac{1+x}{1+x} - 1 \\ = \ln(1+x) \end{aligned}$$

$$\therefore \int \ln(1+x) dx = (1+x)\ln(1+x) - x + C$$

(ii) $I_n = \int_0^1 x^n \ln(1+x) dx$

$$= \int_0^1 x^n d[(1+x) \ln(1+x) - x]$$

$$= \left[x^n [(1+x) \ln(1+x) - x] \right]_0^1 - n \int_0^1 x^{n-1} [(1+x) \ln(1+x) - x] dx$$

$$= (2 \ln 2 - 1) - n \int_0^1 x^{n-1} \ln(1+x) dx - n \int_0^1 x^n \ln(1+x) dx + n \int_0^1 x^n dx \quad \text{①}$$

$$= 2 \ln 2 - 1 - n I_{n-1} - n I_n + n \left[\frac{x^{n+1}}{n+1} \right]_0^1$$

$$= 2 \ln 2 - 1 - n I_{n-1} - n I_n + \frac{n}{n+1}$$

$$\begin{aligned} \therefore (n+1) I_n &= 2 \ln 2 - n I_{n-1} - 1 + \frac{n}{n+1} \\ &= 2 \ln 2 - \frac{1}{n+1} - n I_{n-1} \end{aligned}$$

$$(ii) \quad (n+1)I_n + nI_{n-1} = 2 \ln 2 - \frac{1}{n+1}$$

$$n I_{n-1} + (n-1) I_{n-2} = 2 \ln 2 - \frac{1}{n}$$

$$(n-1)I_{n-2} + (n-2)I_{n-3} = 2 \ln 2 - \frac{1}{n-1}$$

$$(n-2)I_{n-3} + (n-3)I_{n-4} = 2 \ln 2 - \frac{1}{n-2}$$

$$4 I_3 + 3 I_2 = 2 \ln 2 - \frac{1}{4}$$

$$3 I_2 + 2 I_1 = 2 \ln 2 - \frac{1}{3}$$

$$2I_1 + I_0 = 2 \ln 2 - \frac{1}{2}$$

$$(1) - (2) + (3) - (4) + \dots + (n-2) - (n-1) + n$$

$$(\eta+1)I_n + I_0 = -\frac{1}{\eta+1} + \frac{1}{n} - \frac{1}{\eta-1} + \frac{1}{n-2} - \dots - \frac{1}{4} + \frac{1}{3} - \frac{1}{2} + 2 \ln 2$$

$$(\eta+1)I_n + 2\ln 2 - 1 = -\frac{1}{n+1} + \frac{1}{n} - \frac{1}{n-1} + \frac{1}{n-2} - \dots - \frac{1}{4} + \frac{1}{3} - \frac{1}{2} + 2\ln 2$$

$$\therefore (n+1)I_n = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{n} - \frac{1}{n+1}$$

c) (i) $1 + \omega + \omega^2 = 0$

$$\therefore 1 + \omega = -\omega^2$$

$$(1 + \omega)^2 = (-\omega^2)^2$$

$$= \omega^4$$

$$= \omega$$

$$\therefore \omega^3 = 1$$

$$(ii) (1+\omega)^3 = (-\omega^2)^3$$

$$= -\omega^6$$

$$= -(\omega^3)^2$$

$n = 1$

$$(ii) (1+\omega)^{3n} = [(1+\omega)^3]^n$$

$$\text{But } (1+\omega)^{3n} = \binom{3n}{0} + \binom{3n}{1}\omega + \binom{3n}{2}\omega^2 + \binom{3n}{3}\omega^3 + \dots + \binom{3n}{3n}\omega^{3n}$$

$$\therefore (-1)^n = {}^n C_0 + {}^n C_1 \omega + {}^n C_2 \omega^2 + {}^n C_3 + {}^n C_4 \omega + {}^n C_5 \omega^2 + {}^n C_6$$

$$+ \dots + {}^n C_{3n} \quad \boxed{1}$$

Equating real parts

$$(-1)^n = {}^n C_0 + {}^n C_1 \left(-\frac{1}{2}\right) + {}^n C_2 \left(-\frac{1}{2}\right)^2 + {}^n C_3 \left(-\frac{1}{2}\right)^3 + {}^n C_4 \left(-\frac{1}{2}\right)^4 + {}^n C_5 \left(-\frac{1}{2}\right)^5 \quad \text{--- (1)}$$

$$\therefore {}^3C_0 - \frac{1}{2}({}^3C_1 + {}^3C_2) + {}^3C_3 - \frac{1}{2}({}^3C_4 + {}^3C_5) + {}^3C_6 - \dots + {}^3C_n = (-1)^n$$

P.14